

**Response to the Editor:**

Dear Editor,

Thank you for your consideration of our manuscript, “Development and internal validation of early- and mid-pregnancy prediction models for macrosomia in nulliparous singleton pregnancies.” We appreciate the opportunity to revise the manuscript and are grateful for the constructive comments from the reviewers and the editor.

We have carefully revised the manuscript and addressed all questions point by point. In particular, we clarified the scope of the study as model development with internal validation, added a limited external evaluation using an independent Dutch cohort with harmonizable predictors, included gestational-age-adjusted LGA sensitivity analyses, expanded the reporting of calibration and threshold-based operating characteristics, added a selection-bias assessment, and strengthened the discussion of limitations and exploratory findings.

We have uploaded the following revised files: a revised manuscript with the updated sections highlighted in red; a clean version of the revised manuscript; a point-by-point response to the reviewers and updated Supplementary Materials.

We hope that the revised manuscript is now suitable for further consideration in *Archives of Gynecology and Obstetrics*.

Best regards,

Xi Long

on behalf of all authors

## Response to the Reviewer:

### Major comments:

Major comment 1. External validation is absent. Internal validation, even with 1,000-iteration repeated cross-validation, cannot replace evaluation on an independent dataset. Models developed on a single US cohort (recruited 2010-2013) carry significant risk of overestimated performance when applied to other populations or eras. The authors acknowledge this in the limitations but the limitation is fundamental for a prediction modeling paper in 2026. At minimum, the authors should perform a temporal or site-stratified split-cohort sensitivity analysis (e.g., training on 6 of the 8 nuMoM2b clinical sites and validating on the 2 remaining), or apply the model to a public held-out cohort if accessible. This should not be deferred to "future work" given the central claim of clinical utility.

We thank the reviewer for this important comment. We agree that internal validation cannot replace evaluation in an independent population, and that the manuscript should not imply clinical readiness without external validation.

In response, we first revised the manuscript to make the scope of the study more explicit as model development with internal validation. The specific textual changes are as follows:

In the Abstract Conclusion, we added a direct statement on external validation: “*External validation is required before clinical implementation.*” (line 62).

In Section 4.2, Comparison with previous studies, we clarified the intended interpretation of the study: “*The present study therefore provides a clinically oriented prediction framework developed and internally validated in a large prospective cohort, while further external validation is needed before routine clinical implementation.*” (lines 352–354).

In Section 5, Conclusion, we retained a conservative interpretation and removed the exploratory alcohol-related statement. The revised Conclusion states: “*The proposed visit-updated framework may support earlier identification of pregnancies at increased risk, but external validation is needed before clinical use.*” (lines 442–444).

In addition to these textual revisions, and to address the reviewer’s request for independent evaluation, we performed a limited external evaluation using an independent Dutch cohort from Máxima Medical Center in the Netherlands. This cohort included birth outcome data and early-pregnancy maternal demographic and clinical variables but did not include periconceptional dietary indices or serial Visit 2/Visit 3 ultrasound biometry in a form comparable to nuMoM2b. Details of this cohort have been described in our previous work

[1]. Direct external validation of the complete visit-updated models was therefore not feasible.

We consequently evaluated a reduced early-pregnancy model restricted to predictors that could be harmonized across the two cohorts, including maternal age, body mass index, race/ethnicity-related information where available, and gravidity-related information. The detailed results are reported in Supplementary Table S13. Briefly, after removal of non-harmonizable diet variables, the internally validated AUROC of the nuMoM2b Visit-1 model decreased from 0.66 to 0.64 for birthweight >4000 g and from 0.74 to 0.67 for birthweight >4500 g. When the reduced nuMoM2b model was evaluated in the MMC cohort, the AUROCs were 0.60 and 0.64, respectively, which were comparable to the internally validated performance of MMC models developed using the same harmonizable predictor set.

We also added a brief summary of this limited external evaluation to Section 4.4, Strengths and limitations:

*“To further assess cross-cohort transportability within the limits of available external data, we evaluated a reduced early-pregnancy model in an independent Dutch cohort from Máxima Medical Center using predictors that could be harmonized across both cohorts. This analysis was restricted to shared maternal demographic and clinical predictors and did not include periconceptional diet or serial Visit 2/Visit 3 ultrasound measurements. Detailed results are provided in Supplementary Table S13. Overall, the findings supported the potential transferability of the shared early-pregnancy predictor component, but should not be interpreted as external validation of the full diet- and ultrasound-augmented visit-updated models.”* (lines 406–413).

We hope these revisions address the reviewer’s concern by clarifying the intended scope of the internally validated models and by providing an independent external evaluation based on harmonizable predictors.

[1] Wu, Yanqi, et al. "Early prediction of gestational diabetes mellitus using maternal demographic and clinical risk factors." *BMC research notes* 17.1 (2024): 105.

Major comment 2: Macrosomia definition, absolute thresholds versus large-for-gestational-age. The authors use birthweight thresholds >4000 g and >4500 g without considering gestational age at delivery. Modern obstetric guidelines (ISUOG, ACOG, RCOG) increasingly favor LGA defined as >90th percentile birthweight-for-gestational-age,

because absolute thresholds disproportionately reflect gestational age at delivery rather than fetal overgrowth per se. This is particularly relevant given that the cohort includes 7.4% preterm births. The authors should justify the choice of absolute thresholds, run a sensitivity analysis using an LGA definition (e.g., Hadlock or INTERGROWTH-21st >90th percentile birthweight), and discuss concordance between definitions.

We thank the reviewer for this clinically important comment. We agree that absolute birthweight thresholds and large-for-gestational-age definitions capture related but distinct aspects of fetal overgrowth. Birthweight >4000 g and >4500 g were retained as the primary outcomes because they are widely used absolute definitions of macrosomia and directly reflect delivery-related risk associated with absolute fetal size. We also agree that these thresholds do not account for gestational age at delivery, which is particularly relevant given the presence of preterm births in the cohort. We also note that the preterm-birth proportion in the analytic cohort has been corrected to 13.8% (880/6,371, <37 weeks) in the revised manuscript. This correction does not affect any model results.

In response, we added gestational-age-adjusted LGA sensitivity outcomes in Section 2.2, Outcomes:

*“Because absolute birthweight thresholds do not account for gestational age at delivery, we additionally performed sensitivity analyses using large-for-gestational-age birthweight. LGA90 and LGA97 were defined as birthweight above the 90th and 97th percentiles, respectively, for gestational age at delivery and neonatal sex according to the INTERGROWTH-21st Newborn Size Standards. These analyses assessed whether the visit-updated prediction pattern was consistent when fetal overgrowth was defined relative to gestational age rather than by fixed birthweight thresholds.”* This addition appears in Section 2.2, Outcomes (lines 103–108).

The LGA sensitivity analyses were performed using the same visit-updated modeling framework as the primary analysis. We added the corresponding results to Section 3.2:

*“Sensitivity analyses using gestational-age-adjusted LGA outcomes supported the same staged prediction pattern. LGA90 occurred in 1319 pregnancies (20.7%) and LGA97 in 470 pregnancies (7.4%). For LGA90, AUROC increased from 0.61 (95% CI 0.60–0.63) in the early-pregnancy model to 0.72 (0.71–0.74) in the Visit-3 model. For LGA97, AUROC increased from 0.64 (0.61–0.66) to 0.75 (0.72–0.77). AUPRC also improved after incorporation of later ultrasound information. Full discrimination and calibration results for the LGA sensitivity outcomes are provided in Supplementary Table S4, with calibration plots shown in Supplementary Figure S3.”* (lines 278–284).

The full LGA90 and LGA97 sensitivity results, including AUROC, AUPRC, Brier score, calibration intercept, and calibration slope, are now reported in Supplementary Table S4. Calibration plots for these LGA sensitivity outcomes are provided in Supplementary Figure S3.

We also revised Section 4.4, Strengths and limitations to clarify that the two types of outcome definitions should not be interpreted interchangeably:

*“Relatedly, although we added LGA90 and LGA97 sensitivity analyses to account for gestational age at delivery, absolute macrosomia and LGA should not be interpreted interchangeably. Future studies should determine which fetal overgrowth definition best predicts delivery and neonatal complications.”* (lines 420–423).

Major comment 3: Class imbalance handling during model training is not described. The macrosomia >4500 g class represents only 1.2% of pregnancies. The authors report AUPRC, which is appropriate, but the methods do not describe how class imbalance was handled during model training (e.g., class weights, oversampling, undersampling, focal loss, or none). If no rebalancing was performed, the model may systematically underpredict macrosomia and the calibration may suffer at high-probability ranges. Please clarify in Section 2.5 and report calibration intercept and slope, not only the Brier score, ideally stratified by predicted probability deciles.

We thank the reviewer for this important comment. We agree that class imbalance and calibration require explicit reporting, particularly for the rarer birthweight >4500 g outcome.

In response, we revised Section 2.5, Model development and internal validation, to clarify how class imbalance was handled during model training. The revised manuscript now states:

*“Class imbalance was handled using weighted logistic regression, with class weights computed within each cross-validation training fold. The negative class was assigned a weight of 1, and the positive class was weighted by the square root of the negative-to-positive sample ratio to avoid excessive up-weighting of rare positive cases.”* This revision appears in Section 2.5 (lines 164–167).

We also clarified the calibration procedure in the same section:

*“Predicted probabilities were then calibrated using isotonic regression, using a stratified 10% subset of each training fold for calibration before evaluation on the held-out validation fold.”* (lines 167–169).

In addition, we expanded the calibration evaluation beyond the Brier score. Section 2.5 now states:

*“Calibration was assessed using the Brier score, reliability plots, calibration intercept, and calibration slope, and was additionally summarized across predicted-risk calibration blocks.” (lines 172–174).*

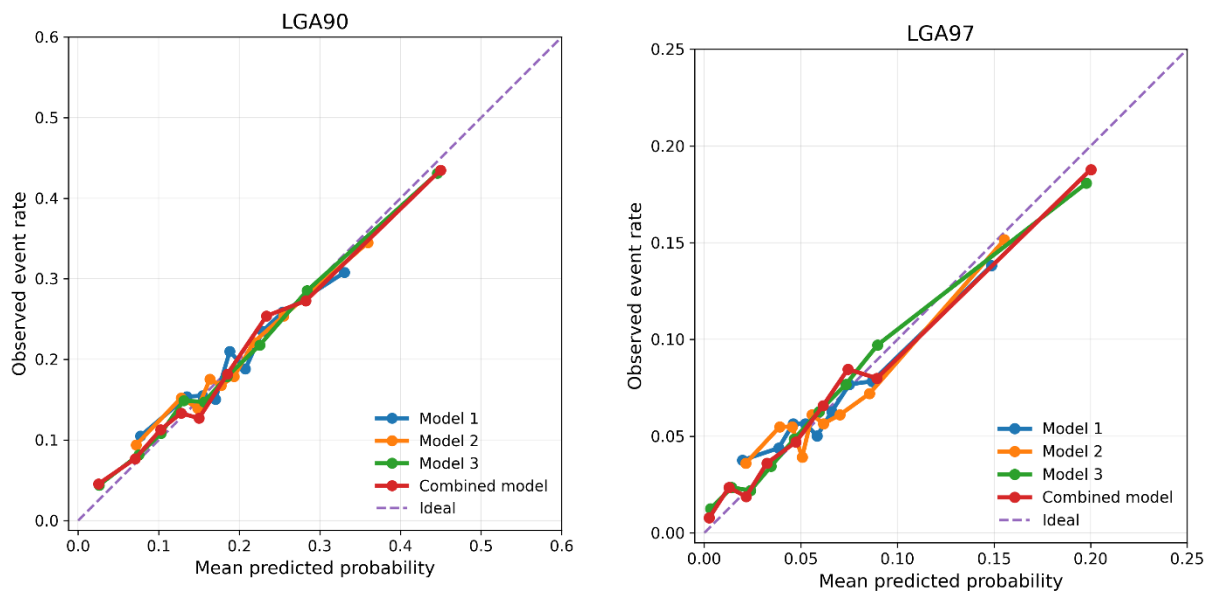
The corresponding performance and calibration results are now reported in the Supplementary Material. For the primary macrosomia outcomes, additional performance metrics, including AUPRC and Brier score, are provided in Supplementary Table S5, together with calibration intercept and calibration slope. Calibration plots for the primary outcomes are provided in Supplementary Figure S2. To make these additional results visible in the main Results without repeating the AUROC results already reported in Section 3.2, we added the following summary:

*“Additional performance metrics for the final models are provided in Supplementary Table S5 and Supplementary Figure S2. For the Combined model, the AUPRC was 0.21 (95% CI 0.19–0.24) and the Brier score was 0.07 (0.06–0.07) for birthweight >4000 g. For birthweight >4500 g, the corresponding values were 0.07 (0.04–0.12) and 0.01 (0.01–0.01). Supplementary Table S5 also reports calibration intercepts and slopes, and Supplementary Figure S2 shows the corresponding calibration plots.” (lines 259–263).*

For the LGA90 and LGA97 sensitivity outcomes, corresponding discrimination and calibration metrics are provided in Supplementary Table S4, and calibration plots are shown in Supplementary Figure S3.

These additions clarify that class imbalance was handled during model training using fold-specific class weights, and that calibration was evaluated using multiple complementary measures rather than the Brier score alone.

**Supplementary Figure S3. Calibration plots for LGA sensitivity outcomes.** Calibration plots were generated from calibrated out-of-fold predicted probabilities from repeated 10-fold cross-validation. Each point represents one predicted-risk block, plotted as the mean predicted probability against the observed event rate. The dashed diagonal line indicates perfect calibration. Panels show sensitivity outcomes defined as birthweight above the INTERGROWTH-21st gestational-age- and sex-specific (a) 90th percentile and (b) 97th percentile. Model 1, Model 2, Model 3, and the Combined model are shown separately.



**Supplementary Table S4. Discrimination and calibration of class-weighted isotonic-calibrated models for LGA sensitivity outcomes.**

| Outcome                                               | Model          | Event rate | AUROC (95% CI)      | AUPRC (95% CI)      | Brier score (95% CI) | Calibration intercept (95% CI) | Calibration slope (95% CI) |
|-------------------------------------------------------|----------------|------------|---------------------|---------------------|----------------------|--------------------------------|----------------------------|
| INTERGROWTH-21st birthweight centile >90th percentile | Model 1        | 20.7%      | 0.613 (0.596–0.629) | 0.279 (0.260–0.300) | 0.161 (0.155–0.167)  | -0.365 (-0.540 to -0.190)      | 0.718 (0.593–0.841)        |
| INTERGROWTH-21st birthweight centile >90th percentile | Model 2        | 20.7%      | 0.635 (0.618–0.652) | 0.311 (0.289–0.335) | 0.158 (0.152–0.164)  | -0.276 (-0.461 to -0.103)      | 0.784 (0.649–0.911)        |
| INTERGROWTH-21st birthweight centile >90th percentile | Model 3        | 20.7%      | 0.722 (0.707–0.737) | 0.412 (0.386–0.441) | 0.146 (0.141–0.152)  | -0.160 (-0.285 to -0.045)      | 0.864 (0.774–0.949)        |
| INTERGROWTH-21st birthweight centile >90th percentile | Combined model | 20.7%      | 0.727 (0.712–0.742) | 0.412 (0.385–0.439) | 0.146 (0.140–0.151)  | -0.149 (-0.264 to -0.029)      | 0.872 (0.786–0.960)        |

|                                                       |                |      |                        |                        |                        |                           |                        |
|-------------------------------------------------------|----------------|------|------------------------|------------------------|------------------------|---------------------------|------------------------|
| INTERGROWTH-21st birthweight centile >97th percentile | Model 1        | 7.4% | 0.635<br>(0.608–0.662) | 0.127<br>(0.110–0.149) | 0.067<br>(0.062–0.073) | -0.623 (-1.016 to -0.247) | 0.741<br>(0.585–0.893) |
| INTERGROWTH-21st birthweight centile >97th percentile | Model 2        | 7.4% | 0.642<br>(0.614–0.668) | 0.139<br>(0.119–0.166) | 0.067<br>(0.062–0.072) | -0.554 (-0.931 to -0.207) | 0.770<br>(0.629–0.910) |
| INTERGROWTH-21st birthweight centile >97th percentile | Model 3        | 7.4% | 0.746<br>(0.723–0.769) | 0.215<br>(0.186–0.252) | 0.064<br>(0.059–0.068) | -0.384 (-0.681 to -0.088) | 0.824<br>(0.706–0.947) |
| INTERGROWTH-21st birthweight centile >97th percentile | Combined model | 7.4% | 0.751<br>(0.728–0.774) | 0.218<br>(0.187–0.254) | 0.063<br>(0.059–0.068) | -0.367 (-0.654 to -0.109) | 0.832<br>(0.713–0.942) |

**Table note:** LGA sensitivity outcomes were defined using INTERGROWTH-21st birthweight centiles according to gestational age at delivery and neonatal sex. Values are based on calibrated out-of-fold predicted probabilities from repeated 10-fold cross-validation. Class weights were defined using the square root of the negative-to-positive sample ratio, with the negative class assigned a weight of 1. Probability calibration was performed using isotonic regression within the cross-validation framework. Confidence intervals were obtained by bootstrap resampling. AUROC: area under the receiver operating characteristic curve; AUPRC: area under the precision–recall curve; CI: confidence interval; LGA: large for gestational age.

Major comment 4: The alcohol-macrosomia association is biologically implausible and should be either substantiated or removed from the abstract. Maternal alcohol consumption during pregnancy is consistently associated with fetal growth restriction, not macrosomia. The reported adjusted OR 2.76 (95% CI 1.31-5.83) for moderate alcohol and birthweight >4500 g is based on a small subgroup, derived from a multi-step ablation procedure that was not pre-specified, and is most plausibly explained by residual confounding (socioeconomic status, body composition, dietary patterns, recall bias on FFQ). The authors do flag the finding as exploratory and provide caveats in Section 4.3. However, its inclusion in the abstract risks misinterpretation by clinical readers. We recommend either (a) removing the alcohol finding from the abstract entirely, or (b) substantially strengthening the disclaimer language in the abstract, and (c) running a falsification analysis (e.g., negative control outcome such as gestational age at delivery) to test for residual confounding before discussing the finding in clinical-implications terms.

We thank the reviewer for this important comment. We agree that the alcohol-related finding requires particularly cautious interpretation and that its inclusion in the Abstract could lead to overinterpretation by clinical readers.

In response, we removed the alcohol-related finding from the Abstract Results and Abstract Conclusion. The revised Abstract Conclusion now states: “*Routinely available early-pregnancy maternal characteristics and anthropometry provided moderate risk stratification for macrosomia, and later ultrasound measurements improved performance further. External validation is required before clinical implementation.*” (lines 60–62).

We also removed the alcohol-related statement from Section 5, Conclusion. The revised Conclusion states: “*The proposed visit-updated framework may support earlier identification of pregnancies at increased risk, but external validation is needed before clinical use.*” (lines 442–444).

In addition, we revised the presentation of this finding so that it is no longer presented as a clinical-implication result. In the revised manuscript, the alcohol-related finding is retained only in the exploratory dietary component analysis and is discussed as a post-hoc, hypothesis-generating observation.

We also performed a negative-control-type sensitivity analysis using gestational age at delivery as the outcome, as suggested by the reviewer. Using the same alcohol exposure coding and adjustment covariates as in the macrosomia analysis, the alcohol-related variable was not materially associated with gestational age at delivery as a continuous outcome (adjusted  $\beta$  0.099 weeks, 95% CI -0.046 to 0.243). Similar null findings were observed for delivery before 37 weeks (OR 0.92, 95% CI 0.73–1.15) and delivery before 39 weeks (OR 0.90, 95% CI 0.76–1.07). The association with delivery at or after 41 weeks was imprecise and not statistically significant (OR 1.27, 95% CI 0.81–1.94). These results are reported in Supplementary Table S6.

We added the following sentence to Section 3.4, Periconceptional diet component analysis: “*In a negative-control-type sensitivity analysis, the alcohol-related variable was not materially associated with gestational age at delivery or with preterm or early-term delivery (Supplementary Table S6).*” (lines 310–312).

**Supplementary Table S6. Negative-control-type sensitivity analysis of the alcohol-related variable with gestational age at delivery.**

| Outcome                            | Model                      | Estimate              | 95% CI          | Interpretation                                    |
|------------------------------------|----------------------------|-----------------------|-----------------|---------------------------------------------------|
| Gestational age at delivery, weeks | Adjusted linear regression | $\beta$ = 0.099 weeks | -0.046 to 0.243 | No statistically significant material association |

|                    |                              |           |              |                                                   |
|--------------------|------------------------------|-----------|--------------|---------------------------------------------------|
| Delivery <37 weeks | Adjusted logistic regression | OR = 0.92 | 0.73 to 1.15 | No statistically significant material association |
| Delivery <39 weeks | Adjusted logistic regression | OR = 0.90 | 0.76 to 1.07 | No statistically significant material association |
| Delivery ≥41 weeks | Adjusted logistic regression | OR = 1.27 | 0.81 to 1.94 | Imprecise; not statistically significant          |

We also revised Section 4.1 Principal findings to avoid presenting the alcohol-related result as a main clinical finding. The revised text states: “*Exploratory diet-component analyses suggested that most individual dietary components contributed limited incremental predictive information, with one post-hoc alcohol-related finding requiring cautious interpretation.*” (lines 322–324).

We further strengthened the cautionary language in Section 4.2, Comparison with previous studies:

*“The alcohol-related result requires particularly cautious interpretation. Although the Alternative Healthy Eating Index alcoholic drinks component emerged in the exploratory ablation analysis and moderate periconceptional alcohol intake was associated with birthweight >4500 g in adjusted models, this analysis was not designed to test a prespecified causal hypothesis. A negative-control-type sensitivity analysis using gestational age at delivery did not show a material association with the alcohol-related variable, but this does not exclude residual confounding. The finding is also biologically counterintuitive in relation to the broader literature linking maternal alcohol exposure with impaired fetal growth rather than fetal overgrowth. It may therefore reflect residual confounding by socioeconomic status, maternal body composition, broader dietary or lifestyle patterns, or exposure misclassification related to self-reported food-frequency questionnaire data. Accordingly, this result should be regarded as hypothesis-generating only and should not be interpreted as causal or clinically actionable.”* (lines 362–373).

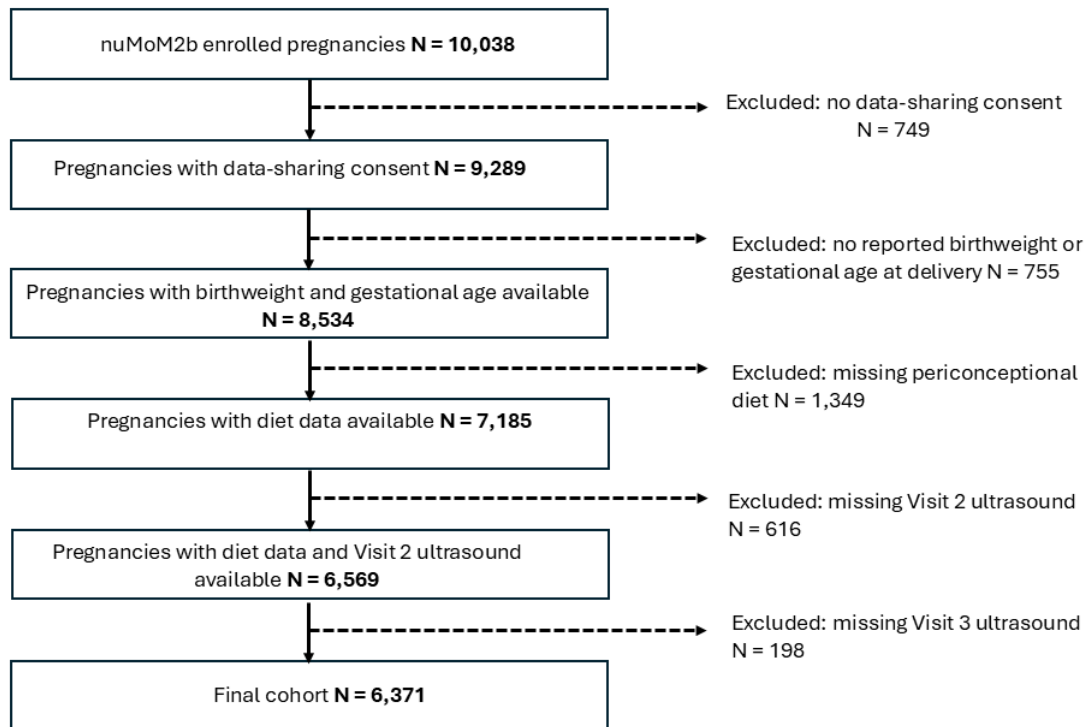
Major comment 5. Selection bias from cohort restriction to women with complete Visit 2 and Visit 3 ultrasound data. The authors excluded women lacking Visit 2 or Visit 3 ultrasound measurements without quantifying the proportion excluded for this reason or characterizing whether excluded women differ systematically from included women (likely they do, with excluded women representing higher-risk pregnancies who delivered preterm or were lost to follow-up). A flow diagram with reasons for exclusion at each step

(CONSORT-like or TRIPOD-AI compliant) and a comparison table of included versus excluded women would substantially strengthen the analysis. This is also a TRIPOD+AI 2024 reporting recommendation.

We thank the reviewer for this important comment. We agree that restricting the analytic cohort to pregnancies with available periconceptional diet data and serial Visit 2/Visit 3 ultrasound fetal biometry data may introduce selection bias and should be reported transparently.

In response, we added a participant flow diagram with sequential, mutually exclusive exclusion steps as Supplementary Figure S4. The flow diagram shows that, among 10,038 women enrolled in nuMoM2b, 749 were excluded because data-sharing consent was unavailable. Among the remaining 9,289 pregnancies, 755 were excluded because birthweight or gestational age at delivery was not reported, leaving 8,534 pregnancies with outcome information. We then excluded 1,349 pregnancies with missing periconceptional diet data, 616 pregnancies with missing Visit 2 U/S fetal biometry data, and 198 additional pregnancies with missing Visit 3 U/S fetal biometry data after Visit 2 fetal biometry data were available. The final analytic cohort included 6,371 pregnancies. Thus, 814 pregnancies were excluded because of incomplete Visit 2 and/or Visit 3 U/S fetal biometry data after diet data were available.

***Supplementary Figure S4. Participant flow for derivation of the analytic cohort.***



We revised Section 2.6, Statistical analysis, to describe this additional selection-bias assessment:

*“Participant flow was summarized using sequential, mutually exclusive exclusion steps. To characterize potential selection during cohort derivation, we compared the final analytic cohort with the two main sequential exclusion groups: pregnancies excluded because of missing periconceptional diet data and pregnancies excluded because of incomplete Visit 2 and/or Visit 3 fetal biometry data after diet data were available. Because these two exclusion mechanisms may reflect different sources of selection, each excluded group was compared separately with the included analytic cohort across maternal sociodemographic characteristics, early-pregnancy clinical characteristics, and available pregnancy outcomes.” (lines 202–209).*

We also revised Section 3.1, Cohort characteristics, to summarize the flow and the included-versus-excluded comparison:

*“Among the 10,038 women enrolled in the nuMoM2b cohort, 6,371 nulliparous pregnancies were included after excluding those lacking consent (N=749), outcome data (N=755), periconceptional diet information (N=1349), or Visit 2/3 ultrasound measurements*

*(N=814), as shown in Supplementary Figure S4. Because complete periconceptional diet and serial ultrasound data were required for the primary analysis, we further compared the included cohort with pregnancies excluded because of missing diet data or incomplete Visit 2/3 ultrasound follow-up (Supplementary Table S7). Excluded pregnancies differed from included pregnancies in several maternal sociodemographic characteristics and pregnancy outcomes, indicating potential selection related to data completeness and follow-up.” (lines 216–223).*

To further characterize potential selection during cohort derivation, we added Supplementary Table S7, which compares the final analytic cohort with pregnancies excluded because of missing periconceptional diet data and pregnancies excluded because of incomplete Visit 2 and/or Visit 3 fetal biometry data after diet data were available. The comparison includes maternal sociodemographic characteristics, preliminary clinical characteristics, and available pregnancy outcomes. The included-versus-excluded comparison showed that excluded pregnancies differed from included pregnancies in several measured characteristics, including maternal age, income, education, smoking before pregnancy, gestational age at delivery, preterm birth, and birthweight. These findings support the reviewer’s concern that data completeness and serial fetal biometry follow-up were not entirely random and may be related to sociodemographic factors, pregnancy course, or follow-up access.

We also revised Section 4.4, Strengths and limitations to state this more explicitly:

*“The requirement for complete periconceptional diet and serial Visit 2/Visit 3 ultrasound data may also have introduced selection bias. Although this restriction was necessary for evaluating visit-updated models incorporating diet and antenatal ultrasound information, the included-versus-excluded comparison suggested that data completeness and follow-up were not entirely random.” (lines 427–430).*

Major comment 6: Decision-threshold operating characteristics are not reported. AUROC summarizes overall discrimination but does not translate directly into clinical utility. The authors should report sensitivity, specificity, PPV, NPV, and decision curve analysis at clinically meaningful probability thresholds (e.g., upper quintile, upper decile of predicted probability). This is essential to allow clinicians to gauge how the model would behave in routine antenatal triage.

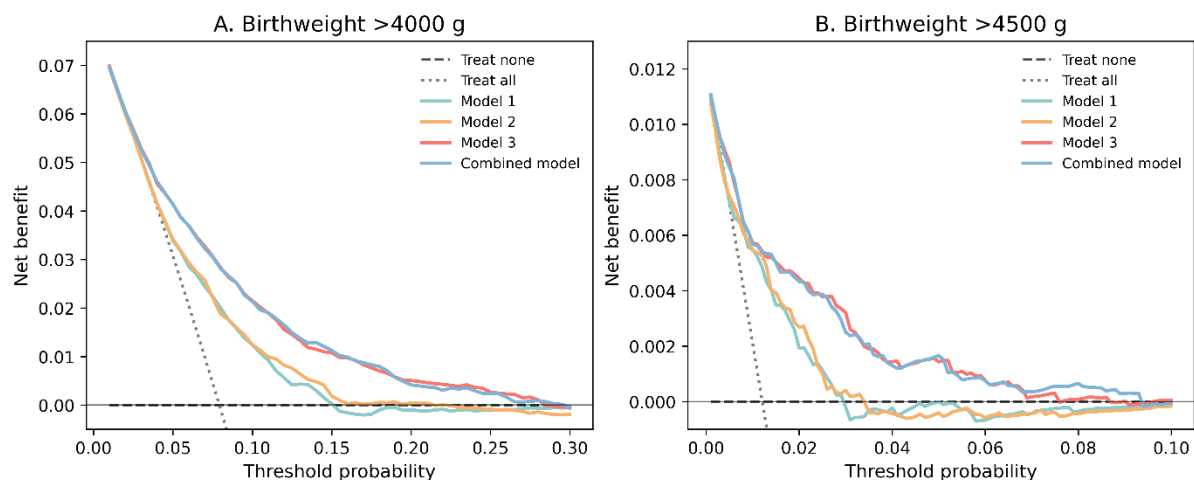
We thank the reviewer for this helpful suggestion. We agree that AUROC alone does not fully describe how the models may behave in clinically interpretable antenatal risk-stratification settings.

In response, we added threshold-based operating characteristics using calibrated out-of-fold predicted probabilities. Because no universally accepted intervention threshold exists for antenatal macrosomia risk stratification, we evaluated the upper quintile and upper decile of predicted risk as prespecified risk-stratification thresholds.

We revised Section 2.5, Model development and internal validation, to state:

*“To provide clinically interpretable operating characteristics, we also evaluated the upper quintile and upper decile of calibrated out-of-fold predicted risk as prespecified risk-stratification thresholds, and calculated the corresponding predicted-risk cutoff, screen-positive proportion, sensitivity, specificity, positive predictive value, and negative predictive value.”* (lines 174–177).

**Supplementary Figure S5. Decision curve analysis for primary macrosomia outcomes.**



The full threshold-based operating characteristics are reported in Supplementary Table S8. We also added decision curve analysis for the two primary macrosomia outcomes as Supplementary Figure S5, using calibrated out-of-fold predicted probabilities to compare model net benefit with treat-all and treat-none reference strategies across threshold-probability ranges.

We also added the following summary to Section 3.2, Model performance and comparison with fetal growth chart–based benchmarks:

*“Threshold-based operating characteristics are shown in Supplementary Table S8. For birthweight >4000 g, the Combined model classified 1,275 pregnancies as screen-positive at the upper-quintile threshold and captured 244 of 505 cases, with a sensitivity of 48.3%, specificity of 82.4%, positive predictive value (PPV) of 19.1%, and negative predictive value (NPV) of 94.9%. At the upper-decile threshold, it captured 155 of 505 cases, with a sensitivity of 30.7%, specificity of 91.8%, PPV of 24.3%, and NPV of 93.9%.*

*For birthweight >4500 g, Model 3 and the Combined model had similar operating characteristics. At the upper-quintile threshold, both captured 51 of 77 cases, with a sensitivity of 66.2%, specificity of 80.6%, PPV of 4.0%, and NPV of 99.5%. At the upper-decile threshold, Model 3 captured 36 of 77 cases, with a sensitivity of 46.8%, specificity of 90.4%, PPV of 5.6%, and NPV of 99.3%; the Combined model captured 35 of 77 cases, with a sensitivity of 45.5%, specificity of 90.4%, PPV of 5.5%, and NPV of 99.3%.” (lines 264–273).*

#### **Minor comments:**

Minor comment 1. The choice of logistic regression as the only modeling approach is reasonable for interpretability, but a brief sensitivity comparison against a non-linear method (random forest, XGBoost, or gradient boosting) in supplementary materials would strengthen the case for or against linearity assumptions.

We thank the reviewer for this helpful suggestion. We agree that a supplementary comparison with non-linear classifiers can help assess whether the primary findings are sensitive to the use of logistic regression as the main modeling framework.

In response, we added a supplementary sensitivity analysis using three commonly used non-linear classifiers: random forest, XGBoost, and gradient boosting. For each primary outcome and each of the four predictor sets, AUROC and AUPRC were estimated from out-of-fold predictions under stratified 10-fold cross-validation. Within each outer training fold, hyperparameters were selected using an inner 3-fold randomized search optimizing average precision. Confidence intervals were estimated using 2,000 bootstrap resamples of the out-of-fold predictions, and class imbalance was handled using the same fold-specific weighting strategy as in the primary analysis.

We revised Section 2.5, Model development and internal validation, to state:

*“As an additional model-form sensitivity analysis, we compared the primary-outcome discrimination results with hyperparameter-tuned random forest, XGBoost, and gradient boosting models, using AUROC and AUPRC from cross-validated out-of-fold predictions.” (lines 186–189).*

The results are reported in Supplementary Table S9. For birthweight >4000 g, the highest non-linear AUROC was 0.745 with XGBoost in Model 3, and the highest AUPRC was 0.215 with random forest in Model 3. For birthweight >4500 g, the highest non-linear AUROC was 0.821 with random forest in the Combined model, and the highest AUPRC was 0.065 with gradient boosting in Model 3.

Because this comparison was intended to evaluate model-form sensitivity rather than to introduce a separate model-selection exercise, we summarized it briefly in the Results section:

*“A separate model-form comparison using random forest, XGBoost, and gradient boosting is reported in Supplementary Table S9. These non-linear classifiers followed the same visit-updated performance pattern and did not show a consistent advantage over logistic regression.”* (lines 285–287).

Overall, the non-linear classifiers provided broadly comparable discrimination but did not show a consistent advantage over the primary logistic-regression framework. We therefore retained logistic regression as the primary modeling approach because of its interpretability, reproducibility, and clinical transparency.

[Minor comment 2. Missing data proportions per predictor are not reported. A supplementary table summarizing missingness for each candidate predictor before k-NN imputation would help readers assess the impact of imputation.](#)

We thank the reviewer for this helpful suggestion. We agree that reporting predictor-level missingness before imputation improves transparency and helps readers assess the potential impact of imputation.

In response, we added Supplementary Table S10, “Predictor-level missingness before k-nearest-neighbor imputation.” This table reports the percentage of missing values for each retained candidate predictor in the final analytic cohort before imputation, using the final analytic sample of 6,371 pregnancies as the denominator.

We also revised Section 2.5, Model development and internal validation, to state:

*“Predictor-level missingness before imputation was summarized in Supplementary Table S10. Missing data among retained candidate predictors were imputed using a k-nearest-neighbor ( $k=5$ ) approach. To prevent information leakage, all preprocessing steps, including imputation and scaling, were performed after data splitting and within the training data only.”* (lines 157–161).

Supplementary Table S10 shows that most retained predictors had low missingness before imputation. Missingness was 16.3% for income level and 12.1% for neck circumference, while most maternal clinical, anthropometric, and fetal biometry predictors had missingness below 10%. These results provide a predictor-level summary of the amount of imputation required in the final analytic cohort.

We further clarified the distinction between cohort-level availability criteria and residual predictor-level missingness. Cohort derivation required availability of Visit 2 and Visit 3 fetal biometry because these measurements formed the core ultrasound component of the visit-updated models. In contrast, Supplementary Table S10 summarizes residual missingness among retained candidate predictors after cohort derivation and before k-nearest-neighbor imputation.

Minor comment 3. Hadlock formula for estimated fetal weight is the standard reference, but it is known to systematically underestimate weight at the upper extremes (>4000 g). This warrants one sentence in the limitations section.

We thank the reviewer for this helpful suggestion. We agree that although the Hadlock formula is widely used for estimated fetal weight, its potential underestimation at the upper end of the fetal weight distribution should be acknowledged when interpreting models and comparators that use EFW.

In response, we revised Section 4.4, Strengths and limitations to state:

*“In addition, because estimated fetal weight was derived using the Hadlock formula, possible underestimation of fetal weight at the upper extremes should be considered when interpreting EFW-based predictors and chart-based comparators.”* (lines 417–420).

Minor comment 4: WHO fetal growth charts are the chosen comparator, but INTERGROWTH-21st and Hadlock-derived percentiles are also widely used in the US. A brief justification of the WHO choice and at minimum a supplementary comparison with INTERGROWTH-21st would improve interpretability for clinical readers.

We thank the reviewer for this helpful comment. We agree that adding an additional fetal growth reference can improve interpretability for clinical readers.

In the original analysis, the WHO fetal growth chart was used as the primary chart-based ultrasound comparator because it provided a standardized reference for deriving ultrasound percentile benchmarks from Visit 2 and Visit 3 fetal biometry. In response to the reviewer’s suggestion, we added a supplementary comparison using INTERGROWTH-21st fetal ultrasound percentiles within the same chart-based comparator framework.

We added the following text to Section 2.4, Fetal growth chart comparator:

*“As a supplementary comparison, we also evaluated INTERGROWTH-21st fetal ultrasound percentiles within the same chart-based comparator framework. These percentiles were*

*assessed analogously to the WHO-derived percentiles as single-predictor ultrasound comparators for subsequent birthweight >4000 g and >4500 g. These chart-based percentiles describe fetal size at the time of ultrasound examination and do not directly project expected birthweight at delivery.” (lines 140–144).*

The corresponding results are reported in Supplementary Table S11, “Comparison of WHO and INTERGROWTH-21st fetal ultrasound percentile comparators.” We also added the following summary to Section 3.2, Model performance and comparison with fetal growth chart-based benchmarks:

*“A supplementary comparison using INTERGROWTH-21st fetal ultrasound percentiles is provided in Supplementary Table S11. The INTERGROWTH-21st comparators showed a similar descriptive pattern to the WHO chart-based comparators, with stronger discrimination for Visit 3 than Visit 2 ultrasound percentiles. For birthweight >4000 g, Visit-3 abdominal circumference and estimated fetal weight percentiles achieved AUROCs of 0.71 and 0.69, respectively. For birthweight >4500 g, the corresponding AUROCs were 0.80 and 0.79. These results were closely aligned with the WHO-based comparator pattern, in which abdominal circumference and estimated fetal weight-related percentiles were among the strongest single ultrasound measures.” (lines 250–258).*

Overall, we hope these additions address the reviewer’s concern by retaining the WHO chart as the primary chart-based comparator while providing an INTERGROWTH-21st supplementary comparison to assess whether the ultrasound benchmark interpretation was sensitive to the fetal growth reference used.

**Minor comment 5. Periconceptional dietary recall via the Block 2005 FFQ is subject to substantial recall bias since it asks about diet 3 months before pregnancy was confirmed. Please mention in limitations.**

We thank the reviewer for this helpful question. We agree that recall bias is an important limitation of the dietary exposure assessment and should be explicitly acknowledged.

In response, we revised Section 4.4, Limitations, to state:

*“Fourth, periconceptional dietary intake was assessed using the Block 2005 Food Frequency Questionnaire, which relied on self-reported recall of usual intake during the 3 months around conception and was therefore subject to recall bias and exposure misclassification. Dietary intake was not assessed longitudinally across pregnancy, limiting inference about changes in diet over time.” (lines 423–427).*

Minor comment 6. The conclusions of the abstract include the alcohol finding. Given its exploratory nature and the biological implausibility, we recommend confining it to the discussion only.

We thank the reviewer for this important question. We agree that the alcohol-related finding was exploratory and should not be emphasized in the Abstract or the Conclusion.

In response, we removed the statements in the Abstract Results and Abstract Conclusion that highlighted the exploratory association between the alcohol-related dietary component and birthweight >4500 g. The revised Abstract Conclusion now focuses only on the overall prediction framework:

*“Routinely available early-pregnancy maternal characteristics and anthropometry provided moderate risk stratification for macrosomia, and later ultrasound measurements improved performance further. External validation is required before clinical implementation.”* (lines 60–62).

We also removed the corresponding alcohol-related statement from the main Conclusion. The revised Section 5, Conclusion now states:

*“The proposed visit-updated framework may support earlier identification of pregnancies at increased risk, but external validation is needed before clinical use.”* (lines 442–444).

In the revised manuscript, the alcohol-related finding is retained only in the Discussion as a post-hoc, hypothesis-generating observation. We strengthened the cautionary interpretation by explicitly noting the biological implausibility of the finding, the potential for residual confounding, and exposure misclassification due to self-reported dietary recall. Therefore, the finding is no longer presented as a main result or conclusion and should not be interpreted as causal or clinically actionable.

Minor comment 7. The introduction mentions postpartum hemorrhage and shoulder dystocia as macrosomia-related outcomes but neither is analyzed. A brief sentence in the discussion on the rationale for limiting outcomes to birthweight thresholds would help.

We thank the reviewer for this helpful suggestion. We agree that the distinction between birthweight-defined fetal overgrowth and downstream delivery or neonatal complications should be made explicit.

In response, we added the following clarification to the section 4.1 Principal findings:

*“The present analysis focused on birthweight-defined fetal overgrowth rather than downstream delivery complications such as postpartum hemorrhage, shoulder dystocia, or birth trauma, which may also depend on delivery management and other clinical factors beyond fetal size alone.” (lines 334–337).*

This clarification explains why the present study used birthweight >4000 g, birthweight >4500 g, and LGA definitions as prediction outcomes, while not analyzing postpartum hemorrhage, shoulder dystocia, or other downstream complications as separate endpoints.

Minor comment 8. Some figures are described in the text without clear cross-reference to the corresponding panel labels (e.g., Figures 5 and 6 ablation analyses). Adding panel labels or explicit subfigure callouts would improve readability.

We thank the reviewer for this helpful suggestion. We agree that clearer figure cross-references improve readability, particularly for the ablation analyses.

In response, we revised the in-text figure references and figure legends to make the correspondence between the text and figures explicit. In Section 3.4, Periconceptional diet component analysis, we now state:

*“Sequential ablation of diet components within Model 1 showed that removal of individual components did not materially change discrimination for birthweight >4000 g (Figure 5), whereas for birthweight >4500 g, removal of the Alternative Healthy Eating Index alcoholic drinks component produced the largest reduction in model performance (Figure 6).” (line 300–304)*

Minor comment 9. Line 11 declares 2875 words for the body, which is at the lower end for this type of methodological paper. We do not request expansion but suggest the authors consider adding short paragraphs on (a) sample size justification for prediction modeling per Riley et al. recommendations, and (b) explicit TRIPOD+AI 2024 checklist coverage.

We thank the reviewer for this constructive suggestion. We agree that adding a brief sample-size context and explicit reporting-guideline coverage improves transparency for this prediction-model study.

In response, we added the following sample-size context statement to Section 2.6, Statistical analysis:

*“Because this secondary analysis used an existing cohort, no a priori sample-size calculation was performed. Following Riley et al. [29], we report the available sample size (N=6,371) and event counts (>4000 g: 505; >4500 g: 77) to contextualize model development and the more cautious interpretation of the rarer >4500 g outcome.”* (lines 191–194).

We also added a completed TRIPOD+AI 2024 checklist to the Supplementary Material as Supplementary Table S12. In Section 2.6, we added:

*“Reporting was checked against the TRIPOD+AI 2024 guidance for clinical prediction model studies [30]; the completed checklist is provided in Supplementary Table S12.”* (lines 209–211).

Minor comment 10. Author Contributions section is truncated mid-sentence (Section 81). Please complete.

We thank the reviewer for noting this issue. We have reviewed and completed the Author Contributions section in the revised manuscript.